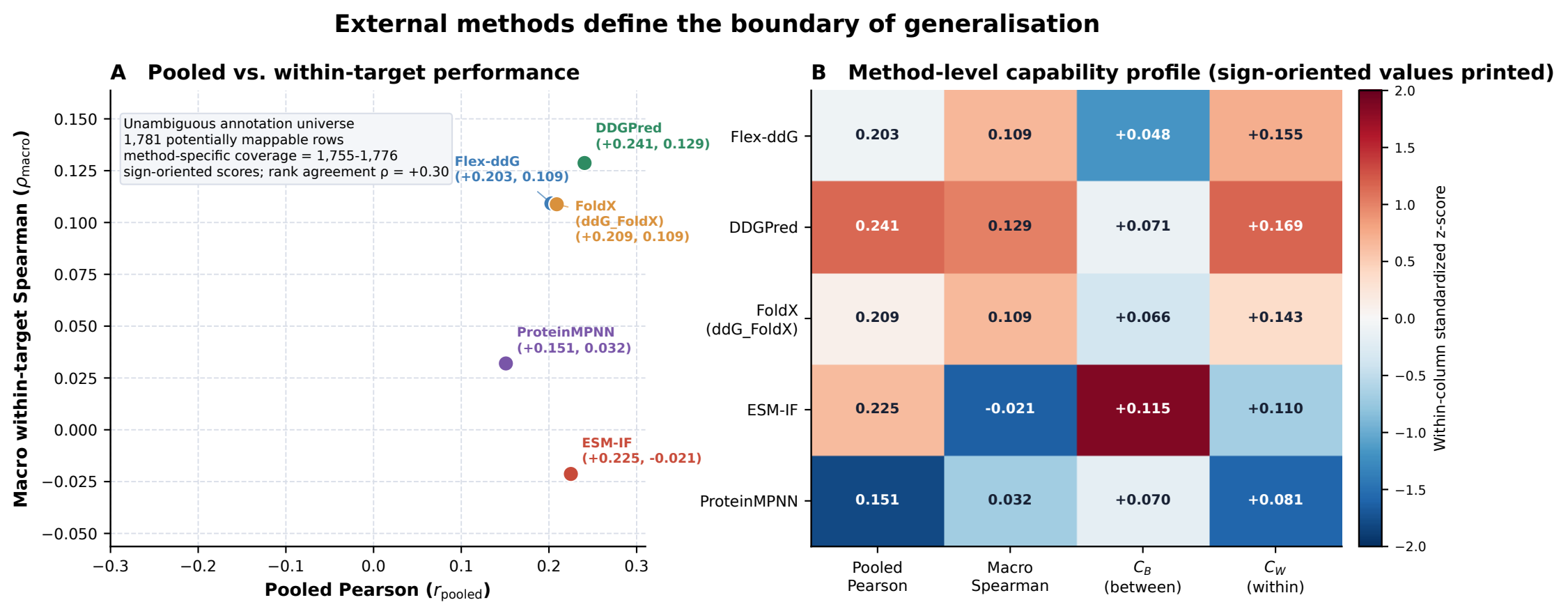
External methods define the boundary of generalisation
A Pooled vs. within-target performance
Unambiguous annotation universe
1,781 potentially mappable rows
method-specific coverage = 1,755-1,776
sign-oriented scores; rank agreement ρ = +0.30
DDGPred
(+0.241, 0.129)
Flex-ddG
(+0.203, 0.109)
FoldX
(ddG_FoldX)
(+0.209, 0.109)
ProteinMPNN
(+0.151, 0.032)
ESM-IF
(+0.225, -0.021)
Macro within-target Spearman (ρ_{macro})
Pooled Pearson (r_{pooled})
0.150
0.125
0.100
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
B Method-level capability profile (sign-oriented values printed)
Flex-ddG
DDGPred
FoldX
(ddG_FoldX)
ESM-IF
ProteinMPNN
0.203
0.109
+0.048
+0.155
0.241
0.129
+0.071
+0.169
0.209
0.109
+0.066
+0.143
0.225
-0.021
+0.115
+0.110
0.151
0.032
+0.070
+0.081
Pooled
Pearson
Macro
Spearman
C_B
(between)
C_W
(within)
Within-column standardized z-score
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0